

Week 1: FinTrust Digital Bank — Understand & Plan

Data Analytics Track Assessment Internship Submission

SECTION 1: General Requirements

A. Understand the Business Problem

1. Problem FinTrust is Trying to Solve: As a growing digital bank, FinTrust struggles with fragmented visibility across customer behavior, transaction operational reliability, risk exposure, and digital support performance. Without unified business intelligence, the bank faces elevated operational friction, high transaction failure/reversal rates, unmitigated fraud risks, and inefficient customer support workflows.

2. Why the Problem is Important: In digital banking, transaction speed, reliability, and security drive customer retention and bottom-line revenue. High transaction failure rates lead to customer churn, while unflagged fraudulent activities lead to financial loss and regulatory penalties.

3. Track Contribution: The Data Analytics track turns raw transactional and demographic data into structured business intelligence. We identify behavioral patterns, evaluate channel efficiency, pinpoint failure/risk bottlenecks, and define key metrics to drive executive decisions.

4. Expected Output: An interactive Executive Data Analytics Dashboard (Power BI/Tableau framework), a comprehensive Data Profiling & Quality Report, a KPI Matrix, and actionable business strategy recommendations.

B. Review of Relevant Resources

1. FinTrust Customer_Data.csv (1,500 records, 12 columns): Contains demographic details (Age, Gender, City, Monthly_Income_Band), customer account profiles (Customer_Segment, Account_Type, Tenure_Months, Account_Status), and Digital_Engagement_Scores. Used for cohort segmentation and churn risk analysis. Limitation: Lacks time-series historical updates for engagement scores or income updates.

2. FinTrust Transaction_Data.csv (12,000 records, 11 columns): Contains granular logs of customer financial activity (Transaction_DateTime, Amount_NGN, Transaction_Type, Channel, Device_Type, Status, Risk_Review_Flag). Used to analyze transactional volume, revenue streams, failure rates, and high-risk activity patterns. Limitation: 96 records contain missing values for Device_Type and Location.

3. FinTrust Data Dictionary.xlsx (12 fields, 7 attributes): Contains explicit metadata, definitions, sample formats, and designated business/modeling use cases for every column. Used to maintain standard metric definitions across tracks.

C. Track Objectives

Objective 1: Clean, profile, and establish relational data models linking customer profiles with transactional behaviors across 12,000 financial logs.

Objective 2: Formulate key performance indicators (KPIs) monitoring transaction success rates, average transaction volume, revenue per segment, and risk exposure flags.

Objective 3: Identify operational bottlenecks, such as channels or locations with disproportionately high failure/reversal rates.

Objective 4: Design and build an interactive 3-tier Executive Dashboard wireframe providing self-service reporting for Product, Operations, and Risk leadership.

D. Success Criteria

Data Completeness & Integrity: 100% of data quality anomalies, missing values, and relationship constraints documented and handled prior to dashboard reporting.

Actionable Analytical Insights: Identification of specific root causes behind failed/reversed transactions and high-risk flags across channels.

Dashboard Usability: An intuitive wireframe with filtering capabilities (by City, Segment, Channel, and Date Range) that enables stakeholders to answer key business questions within 3 clicks.

E. Initial Plan (Weeks 2–4)

Week 2 (Data Preprocessing & Exploratory Data Analysis): Handle missing values in Device_Type and Location; construct foreign-key joints between Customer and Transaction datasets; perform full exploratory analysis across customer segments and transaction channels.

Week 3 (KPI Calculation & Data Modeling): Build aggregated DAX/SQL measures for adoption rates, failure rates, total volume, and risk ratios; establish star-schema data modeling in Power BI.

Week 4 (Dashboard Build, Integration & Final Delivery): Construct final dashboard visual components, implement user-interactivity filters, perform cross-track integration testing, and deliver executive insights deck.

SECTION 2: DATA ANALYTICS TRACK — Specific Deliverables

Part A: Business Understanding

1. Core Business Questions for Management:

- Which transaction channels (Mobile App, Web, ATM, POS, USSD) deliver the highest success rates versus failure/reversal rates?
- What is the transactional lifetime value and average spend across different customer segments (Premium, SME, Everyday, Student)?
- Which locations or channels exhibit disproportionate concentrations of Risk_Review_Flag = Yes?
- How does digital engagement score correlate with account status (Active, Dormant, Restricted)?

2. Business Decisions Supported by Data Analysis:

- **Resource Allocation:** Prioritizing engineering fixes on channels with high failure rates (e.g., USSD or ATM integrations).
- **Risk & Security Enforcement:** Step-up authentication or limit adjustments for specific geographic/channel combinations showing elevated risk flags.
- **Customer Retention:** Targeted loyalty programs or feature adoption nudges for low-engagement customer segments.

3. Benefiting Stakeholders:

- **Executive Leadership (C-Suite):** For strategic growth and high-level financial planning.
- **Operations & Engineering Team:** To monitor system uptime, gateway health, and transaction processing reliability.
- **Risk & Compliance Team:** To investigate flagged high-risk patterns and reduce fraudulent exposure.

- **Product & Marketing Team:** To optimize customer segment engagement and feature rollout.

Part B: Data Understanding & Profiling Report

1. Customer Dataset Structure (Customer_data)

- **Record Count:** 1,500 rows | **Column Count:** 12 columns
- **Primary Key:** Customer_ID (100% unique, format FT-C00001 - FT-C01500)
- **Missing Values:** 0 missing values across all columns.

Column Name	Data Type	Data Category	Values / Range / Unique Counts	Business Notes
Customer_ID	String	Identifier	1,500 unique values	Primary key
Customer_Name	String	Categorical	180 distinct names	Synthetic labels
Age	Integer	Numerical	Min: 18, Max: 65, Mean: 41.59	Adult demographic profile
Gender	String	Categorical	Male (727), Female (722), Prefer not to say (51)	Balanced gender split
City	String	Categorical	8 Major Nigerian Cities (Lagos, Kano, Abuja, etc.)	Geographic distribution
Customer_Segment	String	Categorical	Everyday (711), Premium (295), Student (278), SME (216)	Market tier
Account_Type	String	Categorical	Savings (893), Current (378), Premium (229)	Account structure
Tenure_Months	Integer	Numerical	Min: 1, Max: 96 months	Customer longevity

Digital_Engagement_Score	Float	Numerical	Min: 16.7, Max: 100	Adoption metric
Monthly_Income_Band	String	Categorical	5 Bands (<100k, 100k- 249k, 250k- 499k, 500k- 999k, 1m+)	Wealth indicator
Preferred_Channel	String	Categorical	Mobile App, Web, ATM, POS, USSD	Preferred interface
Account_Status	String	Categorical	Active (1,367), Dormant (107), Restricted (26)	Lifecycle state

2. Transaction Dataset Structure (Transaction_data)

- **Record Count:** 12,000 rows | **Column Count:** 11 columns
- **Primary Key:** Transaction_ID (100% unique, format FT-T0000001 - FT-T012000)
- **Foreign Key:** Customer_ID (Links 100% cleanly to Customer_data with zero orphaned keys).
- **Date Span:** January 1, 2026 to December 3, 2026.

Column Name,	Data Type,	Data Category,	Values / Range / Unique Counts	Business Notes
Transaction_ID	String	Identifier	12,000 unique values	Unique transaction key
Customer_ID	String	Identifier / FK	1,500 unique customer references	Relational key
Transaction_DateTime	DateTime	Temporal	2026-01-01 to 2026-12-03	Full year timestamps
Transaction_Type	String	Categorical	Transfer (3549), Card Purchase (3033), Bill Pay (1475), Deposit (1328), Cash Withdrawal (1430) etc.	Core activities
Amount_NGN	Float	Numerical	Min: ₦100.17, Max:	Financial volume

			₦693,454.45	
Channel	String	Categorical	Mobile App (5102), POS (2393), Web (1869), ATM (1747), USSD	Operational delivery
Device_Type	String	Categorical	Android (4797), iOS (2920), POS, Web, ATM, 96 Nulls	Endpoint hardware
Location	String	Categorical	8 Nigerian Cities, 96 Nulls	Geographic point
International_Transaction	String	Categorical	No (11,412), Yes (588)	Cross-border tag
Transaction_Status	String	Categorical	Successful (10856), Failed (630), Reversed (326), Pending (188)	Operational health
Risk_Review_Flag	String	Categorical	No (9648), Yes (2352)	Risk review indicator

3. Relational Mapping & Data Quality Observations

- **Entity-Relationship:** 1-to-Many (1: N) relationship between Customer_data.Customer_ID and Transaction_data.Customer_ID. Average of 8 transactions per customer.
- **Missing Data Quality Issues:** 96 transaction records (0.8%) have missing values (NaN) in both Device_Type and Location simultaneously.
- **Categorical Alignments:** Channel names in transactions match the Preferred Channel categories in customer profiles, enabling variance analysis between preferred vs. actual channel usage.

Part C: Analytical Questions

- **Customer Behavior:** How does average transaction frequency and total volume vary across different Customer Segments and Monthly Income Bands?
- **Channel Performance:** Which transaction channels exhibit the highest failure and reversal rates, and how does this impact user engagement scores?
- **Transaction Status Breakdown:** What proportion of total transaction value (Amount_NGN) is trapped in Failed, Pending, or Reversed states?

- **Risk Profile:** What combination of Location, Channel, and International Status accounts for the highest concentration of Risk_Review_Flag = Yes?
- **Demographic Adoption:** How does digital engagement score vary by customer Age group and City?
- **Cross-Channel Consistency:** How frequently do customers transact on channels outside of their stated Preferred_Channel?

Part D: KPI Planning

KPI Definition	Why It Matters	Required Data
Transaction Success Rate (%) $\frac{\text{Successful Transactions}}{\text{Total Transactions}} \times 100$	Measures overall system reliability and core processing efficiency.	Transaction_Status
Transaction Failure & Reversal Rate (%) $\frac{\text{Failed} + \text{Reversed Txns}}{\text{Total Transactions}} \times 100$	Identifies operational friction, technical glitches, and revenue loss points.	Transaction_Status
Total Gross Transaction Value (GTV) $\sum (\text{Amount_NGN})$	Tracks total monetary liquidity flowing through the bank platform.	Amount_NGN, Transaction_Status
Average Transaction Value (ATV)	Provides baseline spending behavior across customer segments.	Amount_NGN, Customer_Segment

$\frac{\text{Total Amount}}{\text{Total Count}}$		
Risk Review Exposure Rate (%) $\frac{\text{Flagged Risk Txns}}{\text{Total Transactions}} \times 100$	Quantifies overall potential fraud exposure requiring compliance review.	Risk_Review_Flag
Active Customer Ratio (%) $\frac{\text{Active Customers}}{\text{Total Customers}} \times 100$	Evaluates customer retention health and portfolio churn risk.	Account_Status
Channel Load Distribution (%) $\frac{\text{Txns per Channel}}{\text{Total Txns}} \times 100$	Guides infrastructure scaling based on customer channel preferences.	Channel

Part E: Dashboard Planning & Wireframe

1. Layout Architecture

- **Top Header Filter Bar:** City Slicer | Customer Segment Slicer | Channel Slicer | Date Range Picker.
- **Row 1: Executive KPI Cards (4 Summary Tiles)**
 - Total Volume (NGN) | Total Transactions Count | Transaction Success Rate (%) | High-Risk Flag Count
- **Row 2: Operational Reliability Analysis (Middle Section)**

- Chart 1 (Bar Chart): Transaction Status Breakdown by Processing Channel (Mobile, Web, ATM, POS, USSD).
- Chart 2 (Line Chart): Daily/Monthly Transaction Volume & Failure Trends over 2026.
- **Row 3: Customer & Risk Demographics (Bottom Section)**
 - Chart 3 (Donut Chart): Transaction Value Distribution by Customer Segment (Premium, SME, Everyday, Student).
 - Chart 4 (Heatmap/Bar Chart): Risk Review Flags by Geographic Location and Channel.

2. Key Insights Expected to Communicate

- Operational bottleneck identification by showing which specific channel drives the highest failure rates.
- Segmentation analysis showing which demographic tier contributes the highest net fee volume.
- Geographic risk concentration map to assist risk teams in targeting verification protocols.

SECTION 3 & PROFESSIONAL DELIVERABLES

Future Dependencies (Planning for Weeks 2–4)

- **Dependency to Data Science Track:** Data Analytics will supply baseline segment statistics and high-risk channel profiles to assist DS feature engineering.
- **Dependency to ML Engineering Track:** Schema constraints and data quality flags identified in DA will inform MLE pipeline validation logic.
- **Dependency from Project Management Track:** Timeline alignment for dashboard review milestones.